

LESSON 4

If...Then & Biconditionals

Name: _____ Date: _____

Conditionals — “if...then”

A **conditional** has the form **if p , then q** , written $p \rightarrow q$.

- p = **hypothesis** (antecedent) — the “if” part.
- q = **conclusion** (consequent) — the “then” part.

PROMISE METAPHOR

“If you finish your chores, then you get dessert.” Think of $p \rightarrow q$ as a promise. The promise is broken only when chores are done (p true) but dessert is denied (q false).

If chores are not done (p false), the promise is not broken — dessert might still happen for another reason, and the “if...then” can still count as true in logic.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Only one false row: $p \rightarrow q$ is false exactly when a true hypothesis leads to a false conclusion.
Memorize that row: $T \rightarrow F = F$.

Converse, inverse, contrapositive

Start with: “If a number is divisible by 4, then it is even.” Let p = divisible by 4, q = even.

Form	Symbol	Sentence	True?
Original	$p \rightarrow q$	If divisible by 4, then even.	Yes
Converse	$q \rightarrow p$	If even, then divisible by 4.	No (6)
Inverse	$\neg p \rightarrow \neg q$	If not div. by 4, then not even.	No (6)
Contrapositive	$\neg q \rightarrow \neg p$	If not even, then not div. by 4.	Yes

BIG TAKEAWAY

A conditional and its

CONTRAPOSITIVE

always have the same truth value. The converse and inverse need their own checks — do not assume they match the original.

Biconditional — “if and only if”

$p \leftrightarrow q$ means both $p \rightarrow q$ and $q \rightarrow p$. Both directions must hold.

EXAMPLE

“A triangle is equilateral if and only if it has three equal angles.” Being equilateral and having three equal angles are equivalent — each implies the other.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Notice: $\leftrightarrow$ is true when both parts match (both T or both F).

Practice — Lesson 4

4.1 WARM-UP

Identify hypothesis and conclusion, then write symbols.

1. If it snows, then school is canceled.
2. If you press the power button, then the laptop turns on.

P , Q , AND $P \rightarrow Q$

4.2 WARM-UP

Let $p = \mathbf{F}$, $q = \mathbf{T}$. Find the truth value of $p \rightarrow q$ and $p \leftrightarrow q$.

TWO ANSWERS + ONE-SENTENCE REASON FOR $\rightarrow$

4.3 STANDARD

Original: "If a shape is a square, then it has four sides." Write the converse, inverse, and contrapositive in English. Label each true or false.

THREE SENTENCES + T/F

4.4 STANDARD

Build a truth table for $p \rightarrow q$. Circle the only row where the conditional is false. Explain that row in the promise metaphor.

TABLE + EXPLANATION

4.5 STANDARD

Which of these English sentences are biconditionals? Rewrite each biconditional as $p \leftrightarrow q$.

1. You may vote if and only if you are 18 or older.
2. If you study, then you learn.
3. A number is odd if and only if it is not divisible by 2.

YOUR ANSWERS

4.6 CHALLENGE

Original: “If a student earns at least 90%, then they get an A.” Give a realistic **false converse** scenario (converse is false but sounds plausible at first). Then write the contrapositive and explain why it matches the original’s truth.

SCENARIO + CONTRAPOSITIVE + EXPLANATION

4.7 CHALLENGE

Build a full truth table for $(p \rightarrow q) \leftrightarrow (q \rightarrow p)$. When are a conditional and its converse both true at the same time? Relate your answer to $\leftrightarrow$.

TABLE + DISCUSSION

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \leftrightarrow (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T